

Chapter 9: Evaluation Techniques

HCI — Human Computer Interaction (Dix et al., 3rd Ed.) | Exam Short Notes

9.1 What is Evaluation?

Core Idea: Evaluation assesses designs and tests systems to ensure they behave as expected and meet user requirements. It should occur **throughout the design life cycle**, not just at the end.

- Early evaluation is preferred — problems are **cheaper and easier to fix** at early stages.
- There is a close link between evaluation and **prototyping techniques**.
- Two broad headings:
 - **Expert analysis** — designer/usability expert evaluates without direct user involvement; useful for early designs and prototypes.
 - **User participation** — studies actual use of the system; requires at least a working prototype.

9.2 Goals of Evaluation

Three Main Goals:

1. Assess the **extent and accessibility of system functionality**
2. Assess **users' experience of the interaction**
3. Identify **specific problems** with the system

- **Functionality:** Does the system enable users to perform intended tasks? Includes matching user expectations (e.g., a filing clerk expects postal-address-based retrieval).
- **User Experience:** Ease of learning, usability, satisfaction, enjoyment, emotional response; identify areas that overload the user.
- **Specific Problems:** Identify trouble-spots that cause unexpected results or confusion, then rectify them.

9.3 Evaluation Through Expert Analysis

Expert analysis = Designer or human factors expert assesses the design's impact on a typical user. Identifies areas violating **cognitive principles** or ignoring **empirical results**.

Advantages: Cheap, flexible, usable at any stage (spec → storyboard → prototype → final).

Limitation: Does *not* assess actual use — only checks upholding of usability principles.

Four Approaches to Expert Analysis:

1. Cognitive Walkthrough
2. Heuristic Evaluation
3. Model-Based Evaluation
4. Using Previous Studies

9.3.1 Cognitive Walkthrough

Proposed by **Polson et al.** — introduces psychological theory into the subjective walkthrough. Origin: *code walkthrough* from software engineering.

Main Focus: How easy is the system to learn — specifically **learning through exploration**.

Four Prerequisites (what you need):

1. A specification or prototype of the system (fairly detailed)
2. A description of the task (representative)
3. A complete written list of actions to complete the task
4. An indication of who the users are (experience/knowledge)

Four Questions Asked Per Action Step:

1. Is the **effect of the action** the same as the **user's goal** at that point?
2. Will users **see** that the action is available?
3. Once found, will users **know** it is the one they need?

4. After the action, will users **understand the feedback** they get?

Documentation:

- Produce standard **cover form** (task details, evaluator names, date).
- Separate **action form** per step answering 4 questions.
- **Usability problem report sheet** for every negative answer — include severity and likelihood.

9.3.2 Heuristic Evaluation

Developed by **Jakob Nielsen and Rolf Molich**. Critique a system using a set of simple, general heuristics. Often called a "**discount usability**" technique.

Evaluators: 3–5 evaluators independently; 5 evaluators $\approx$ **75%** of all usability problems found.

Severity Rating Scale (0–4):

- **0** — Not a usability problem
- **1** — Cosmetic only; fix if time permits
- **2** — Minor; low priority fix
- **3** — Major; high priority fix
- **4** — Usability catastrophe; must fix before release

Nielsen's 10 Heuristics:

1. **Visibility of system status** — keep users informed via timely feedback
2. **Match between system and real world** — use user's language; natural/logical order
3. **User control and freedom** — support undo/redo; provide emergency exit
4. **Consistency and standards** — same words/actions mean same things; follow platform conventions
5. **Error prevention** — design to prevent errors, not just handle them
6. **Recognition rather than recall** — make objects/options visible; minimize memory load
7. **Flexibility and efficiency of use** — accelerators for expert users; tailor frequent actions
8. **Aesthetic and minimalist design** — no irrelevant information; every extra unit competes with relevant ones
9. **Help users recognize, diagnose, and recover from errors** — plain language, precise problem description, constructive solution
10. **Help and documentation** — easy to search, task-focused, lists concrete steps

9.3.3 Model-Based Evaluation

- **GOMS** (Goals, Operators, Methods, Selection) — predicts user performance; filters design options.
- **Keystroke-Level Model** — predicts time for low-level physical tasks.
- **Design Rationale** (Ch.6) — evaluates design options against criteria.
- **Dialog Models / State Transition Networks** — check for unreachable states, circular dialogs, complexity.

9.3.4 Using Previous Studies in Evaluation

- Use existing experimental results as evidence to support or refute design decisions.
- Avoids costly repetition of experiments.
- **Caution:** Results don't hold arbitrarily across contexts — reviewer must note experimental design, participants, and assumptions.
- Requires expert knowledge to apply correctly.

9.4 Evaluation Through User Participation

Expert techniques are not a replacement for testing with **actual users**. User participation typically occurs at **later stages** when at least a working prototype exists.

Methods: Empirical, Observational, Query, Physiological Monitoring.

9.4.1 Styles of Evaluation

Laboratory Studies

- Controlled environment; sophisticated AV equipment, two-way mirrors.
- Interruption-free but **lacks real-world context**.
- Good for: dangerous locations, controlled comparisons, single-user constrained tasks.

Field Studies

- Evaluator goes into the user's **real work environment**.
- Retains context; captures real interactions between people/systems.
- Difficult: noise, interruptions, movement.
- **Preferred overall** — studies interaction as it actually occurs.

9.4.2 Empirical Methods: Experimental Evaluation

Controlled experiment — provides **empirical evidence** to support/refute a hypothesis. Most powerful evaluation method.

Key Components:**Participants:**

- Must match the **expected user population** as closely as possible.
- Nielsen & Landauer: 1 participant finds $\approx 1/3$ of problems; little gain beyond 5 for observational studies.
- For controlled experiments with statistical analysis: **at least 10 participants** recommended.

Variables:

- **Independent Variable (IV):** What is manipulated (e.g., interface style, number of menu items, icon design). Each value = a *level*.
- **Dependent Variable (DV):** What is measured (e.g., task completion time, number of errors, user preference).

Hypotheses:

- A prediction of outcome in terms of IV and DV.
- Goal: disprove the **null hypothesis** (no difference exists).
- Significant result = differences would not have occurred by chance.

Experimental Design:

Design		How it works	Trade-off
Between-subjects (randomized)	(ran-	Each participant in one condition only	Controls learning effects; needs more participants
Within-subjects (repeated measures)	(re-	Each participant in all conditions	Fewer participants; risk of learning transfer
Mixed design		One IV between-groups, one within-groups	Compromise approach

Statistical Measures:

- First two rules: **look at the data** and **save the data**.
- Watch for **outliers** (transcription errors or freak events).
- **Parametric tests** — assume normal distribution; more powerful (e.g., Student's *t*-test, ANOVA, regression).
- **Non-parametric tests** — no distribution assumption; rank-based; more robust to outliers but less powerful (e.g., Wilcoxon/Mann-Whitney, Spearman's rank correlation).
- **Contingency tests** — for discrete data (e.g., chi-squared).

Three Statistical Questions:

1. **Is there a difference?** (Hypothesis testing — e.g., "99% certain menu A is faster")
2. **How big is the difference?** (Point estimation — averages)
3. **How accurate is the estimate?** (Standard deviation / confidence intervals)

Statistical Test Selection Guide (Table 9.1):

Test type	IV	DV / Test
Parametric	Two valued	Student's <i>t</i> -test
	Discrete	ANOVA
	Continuous	Linear regression
Non-parametric	Two valued	Wilcoxon / Mann-Whitney
	Discrete	Rank-sum ANOVA
	Continuous	Spearman's rank correlation
Contingency	Discrete	Chi-squared test

Studies of Groups of Users:

- Group experiments are **far more complex** — more participants, longer sessions, harder to control variation.
- Data gathering: multiple video streams, keystroke logs, synchronization challenge.
- Analysis options: within-group design, micro-analysis, or **anecdotal/critical incident analysis**.
- **Ethnography** (from social sciences) — detailed recording of real interactions without imposing a viewpoint; discussed more in Ch.13.

9.4.3 Observational Techniques

Think Aloud:

- User talks through what they are doing, why, and what they believe is happening.
- **Advantage:** Simple; no special expertise; reveals actual use; usable throughout design.
- **Disadvantage:** Information is subjective; observation may alter behavior; like "the centipede who was asked how he walked."

Cooperative Evaluation:

- Variation of think aloud — user is treated as a **collaborator**, not just a participant.
- Evaluator can ask "why?" or "what if?"; user can ask for clarification.
- Less constrained; user is encouraged to criticize; evaluator can clarify confusions immediately.

Protocol Analysis — Recording Methods:

Method	Description	Issue
Paper & Pencil	Cheap; allows interpretations	Limited by writing speed
Audio Recording	Useful with think aloud	Hard to match to exact actions
Video Recording	Shows what participant does	Camera positioning; transcription
Computer Logging	Automatic; unobtrusive; good for longitudinal studies	Semantics-free; volume of data
User Notebooks	Participants log own activity	Coarse level; interpreted records

Post-Task Walkthroughs:

- Transcript is replayed to participant **after the task** — invited to comment or explain actions.
- Done immediately (fresher memory) or after a delay (analyst frames questions better).
- Essential when think aloud is impossible (e.g., during intensive or critical tasks).

9.4.4 Query Techniques

Philosophy: **"Ask the user"** — get user viewpoint directly. Simple and cheap. However, information is necessarily **subjective** and may be a rationalized account.
Two main types: **Interviews** and **Questionnaires**.

Interviews:

- Direct, structured information gathering; evaluator can **probe deeply**.
- Top-down approach: general → leading questions ("why?" / "what if?").
- Effective for high-level evaluation, preferences, impressions; reveals unanticipated problems.
- Should be **planned in advance** with central questions; adapted per user.

Questionnaires:

- Less flexible than interviews; questions fixed in advance; but **wider reach**, faster, more rigorous analysis.
- Must be **well designed**; pilot test on 4–5 users before release.
- Return rate is typically **25–30%**; distribute to random subset.

Question Types in Questionnaires:

Type	Description	Use
General	Background info (age, occupation, experience)	User profiling
Open-ended	User's own unprompted opinion	Subjective; hard to analyze
Scalar	Rate agreement on numeric scale (e.g., 1–5)	Measuring attitudes
Multi-choice	Select from given options	Previous experience
Ranked	Order items by preference	User priorities

- **Scale advice:** 1–5 or 1–7 scales are preferred. Odd-numbered scales allow neutral option; even-numbered scales force a side.
- Prefer **closed questions** (scalar, ranked, multi-choice) to reduce respondent effort and ease analysis.

9.4.5 Evaluation Through Monitoring Physiological Responses

"Objective usability testing" — measures physiological responses directly rather than relying on self-reporting.

Eye Tracking:

- Eye movements reflect amount of **cognitive processing** required.
- Measurements: **number of fixations** (more = less efficient search), **fixation duration** (longer = difficulty), **scan path** (shows cognitive load and search strategy).
- Equipment is still expensive; promising for website design analysis.

Physiological Measurements:

- **Heart activity** (blood pressure, pulse) — responds to stress/anger.
- **Sweat glands / GSR** (galvanic skin response) — indicates arousal and mental effort.
- **EMG** (electromyogram) — electrical activity in muscle; reflects task involvement.
- **EEG** (electroencephalogram) — brain activity; associated with decision making, attention, motivation.
- **Limitation:** Unclear relationship between measurements and exact interaction events; still a research area.

9.5 Choosing an Evaluation Method**9.5.1 Eight Factors Distinguishing Evaluation Techniques**

1. **Stage in the design cycle** — design phase vs. implementation phase
2. **Style of evaluation** — laboratory vs. field
3. **Subjectivity vs. objectivity** — interpreter-dependent vs. repeatable results
4. **Type of measures** — qualitative vs. quantitative
5. **Information provided** — low-level vs. high-level
6. **Immediacy of response** — during interaction vs. post-event recall
7. **Intrusiveness** — does the measurement alter behavior?
8. **Resources required** — time, equipment, money, expertise, participant access

Key trade-off: More immediate methods tend to be more intrusive. Objective methods avoid bias but miss unexpected problems. Use **both subjective and objective** approaches when possible.

9.5.2 Classification of Evaluation Techniques**Analytic Techniques (Table 9.4):**

Factor	Cog. Walkthrough	Heuristic Eval.	Review-Based	Model-Based
Stage	Throughout	Throughout	Design	Design
Objective?	No	No	As source	No
Measure	Qualitative	Qualitative	As source	Qualitative
Information	Low level	High level	As source	Low level
Intrusive?	No	No	No	No
Time	Medium	Low	Low-medium	Medium
Expertise	High	Medium	Low	High

Experimental & Query Techniques (Table 9.5):

Factor	Experiment	Interviews	Questionnaire
Stage	Throughout	Throughout	Throughout
Style	Laboratory	Lab/Field	Lab/Field
Objective?	Yes	No	No
Measure	Quantitative	Qual./Quant.	Qual./Quant.
Information	Low/High	High level	High level
Immediacy	Yes	No	No
Intrusive?	Yes	No	No
Time	High	Low	Low
Expertise	Medium	Low	Low

Observational Techniques (Table 9.6):

Factor	Think Aloud	Protocol Analysis	Post-task Walkthrough
Stage	Implementation	Implementation	Implementation
Objective?	No	No	No
Measure	Qualitative	Qualitative	Qualitative
Immediacy	Yes	Yes	No
Intrusive?	Yes	Yes*	No
Time	High	High	Medium
Equipment	Low	High	Low
Expertise	Medium	High	Medium

*Except system logs

Monitoring Techniques (Table 9.7):

Factor	Eye Tracking	Physiological Measurement
Stage	Implementation	Implementation
Style	Lab	Lab
Objective?	Yes	Yes
Measure	Quantitative	Quantitative
Information	Low level	Low level
Immediacy	Yes	Yes
Intrusive?	No**	Yes
Time	Medium/High	Medium/High
Equipment	High	High
Expertise	High	High

**If equipment is not head-mounted

9.6 Summary

- Evaluation is an **integral part of the design process** — throughout the life cycle.
- Aims: test functionality and usability; identify and rectify problems; determine user attitude.
- **Early evaluation** (design stage) — analytic, expert-based, cheap (cognitive walkthrough, heuristics, models, previous work).
- **Later evaluation** (prototype/system stage) — experimental + observational for quantitative/qualitative

results; query techniques for subjective data; physiological monitoring for objective data.

- Evaluation methods vary in: stage, style, subjectivity, measure type, information level, immediacy, intrusiveness, resources.
- **No single best method** — choose based on requirements and available resources.

Expert Analysis	Experimental	Observational	Query
Cognitive Walkthrough	Controlled Experiment	Think Aloud	Interviews
Heuristic Evaluation	Group Studies	Cooperative Evaluation	Questionnaires
Model-Based		Protocol Analysis	
Previous Work		Post-task Walkthrough	
		Eye Tracking	
		Physiological Monitoring	